

Companies must digitally transform engineering processes to be competitive in global markets. Successfully developing products that unite hardware and software requires integrated, adaptive life-cycle solutions that leverage AI, generative AI, ML, model-based systems engineering, and open, versioned digital threads.

Fueling Multimodal Product Development with Systems Engineering and Digital, Adaptive Life-Cycle Solutions

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Questions posed by: IBM

Answers by: Melinda Ballou, Research Director, Agile ALM, Quality and Portfolio Strategies

Q. Products are becoming software-intensive, interconnected systems — "systems of systems." How should companies respond?

A. Organizations must move away from siloed product development to address the demands and complexity of software-intensive, interdependent systems of systems. Coordination is vital to enable complex product development for highly competitive, collaborative, and dynamically changing environments and to address fluctuating compliance and security challenges. Bringing together perspectives and expertise across multiple disciplines for software, electrical, and mechanical engineering demands an overarching approach to systems engineering. Model-based systems engineering (MBSE) is foundational to this approach, which companies increasingly use to enable and drive multidiscipline design, verification, and validation of requirements. MBSE can serve as a basis for coping with increased product complexity and the integration demanded to help support execution for multidisciplinary design, comprehensive engineering life-cycle management strategies, and systems of systems.

High-quality, adaptive software drives competitive positioning and business execution for products as well as coordination with the expanding demands of interactive systems of systems. But when we consider how organizations incorporate software into products, we often still observe disconnects between mechanical, electrical, and/or systems engineering teams and a need to deepen and further evolve software life-cycle processes. Not surprisingly, in these same organizations, software engineering teams as part of IT departments have broader experience and deeper maturity with application development, life-cycle management, and software engineering. Therefore, these companies can benefit by leveraging those teams and MBSE to advance software-intensive product development life-cycle management strategies, which incorporate quality, change management, requirements, and coordination for end-to-end system development.

In engineering environments, the development life cycle needs to span geographically dispersed teams and the partner/supplier ecosystem, embracing emerging technologies and smarter methods to optimize the entire engineering process. This approach is critical for organizations that depend on software for competitive product differentiation across dynamically changing markets.

Q. What are the most promising investments that you see companies making to address increasingly complex product development?

A. In considering product development, it is vital to know what you're seeking to accomplish with the product and the software that's enabling it. Requirements are like the DNA for products. But flexibility and adaptability in how you evolve requirements to be responsive to changing business needs and pressures are imperative. If you don't have the requirements right, and you aren't interacting in adaptive ways to enable change from a business perspective as competitive and other pressures shift, then the product will not reflect what the business needs. This is especially important in rapidly changing, highly competitive markets. Companies need agile, iterative approaches both for gathering requirements and user stories and for scaling those processes to adaptively respond to changing competitive demands across enterprises, suppliers, and customers.

Requirements are becoming increasingly complex as we move toward software-intensive product development. In that context, ML, AI, generative AI (GenAI), and intelligent analytics can have key roles in enabling organizations to benefit from the knowledge of more experienced engineers and prior team experiences. Historical and predictive analytics can help organizations to adapt to changing needs (including compliance demands) and provide visibility into metrics, as examples. These technologies can help identify anomalies and corrections for problematic issues so engineers can more efficiently leverage work that's been done.

AI can augment requirements checking, which frees more sophisticated and experienced engineers for higher-value creative work. Making metrics visible proactively can give engineers creating software and products context so that the data is pragmatic and actionable to improve execution. The rush to adopt GenAI and LLMs to contextualize and inform development over the past 18 months brings significant opportunities. These must be balanced by judicious use, effective governance and guardrails, and tempered expectations. Companies can benefit from vendor-provided LLMs. With effective support (and substantive resourcing), they can build their own contextualized models as needed to differentiate business execution. Both approaches have a role.

There is also the role that model-based systems engineering can play. This approach helps to model requirements and systems design to enable better understanding of required underlying functionality and structured interfaces. When you add simulation to those models, you can help ensure the validity of the design by decomposing the complexity of the product and modeling its subsystems.

Product complexity is also driving the need to coordinate and co-innovate with clients and partners as companies expand their dependence on external suppliers to help develop increasingly sophisticated products. They must consider how they control the product development life cycle, and coordinate both internal and external information, as well as intellectual property and safeguards for proprietary innovations to protect what they're doing.

Therefore, companies require an engineering life-cycle management solution that encompasses partners, suppliers, and customers across dispersed geographies to integrate that feedback into development processes rather than having it be fractured and noncollaborative. A shared engineering system of record (SOR) will enable organizations and external partners and suppliers to have a single source of proof for engineering information and the ability to exchange timely data for reporting, making it easier to integrate development ecosystems with the engineering life cycle.

Q. As products evolve into software-intensive systems that "cannot afford to fail," how should companies adapt development processes that can support the oversight and checks that are demanded in a commercially viable manner at market speed?

A. There must be a shift in development mentality toward "You can't afford to fail" as we look at the life-preserving and product-enabling roles that software now plays. You won't get a second chance in life-critical environments, for instance. So, as you think about "can't afford to fail," what methodologies do you need to embrace to help make software failproof? Government agencies are advocating compliance models, but these fall short if they're an afterthought in the development life cycle for companies building products. Standards published by industry and governmental agencies in the form of compliance models must be baked in up front. AI, GenAI, and ML should also be interwoven into engineering processes, along with governance and oversight. Organizations should establish an integrated, complete engineering life-cycle management solution to help ensure they follow an auditable process so that as compliance standards are incorporated, there's the ability to know what was done to support the regulatory structures in place. MBSE and digital threads can play a role.

As both customers and people experiencing these products, we expect them to perform flawlessly, and we demand that companies automate more and more of the operation of the products we use. Automobiles are a good example of products developed with a "can't afford to fail" methodology — how they've evolved from purely mechanical systems to electromechanical and, now, software-driven systems. Consider the evolution of self-driving automobiles, which are complex systems of systems that cannot afford to fail as they transport families down congested freeways at 65mph. As these products become more software intensive, and for IoT devices that are consuming data, safety-critical development processes are evolving to incorporate the breadth that needs to be encompassed as part of these systems. Organizations must have full transparency into the development process and produce up-to-the-minute status reports to make better decisions throughout the life cycle.

Occasionally organizations will look at what they've done after product completion and say, "Wait a minute. Is this compliant?" Then they backtrack to document their processes and make the product compliant after they've already built it. If compliance isn't incorporated into the product development model underlying the product from the start, organizations may need to rewrite or, at a minimum, do major revisions of the finished product. The engineering teams creating products need to adopt a complete and holistic view through the development cycle for software-intensive products to ensure they meet compliance and regulatory requirements. This can be accomplished only by compliance integration throughout the product life cycle.

Q. Digital engineering is not new, but there seems to be a growing interest in the digital transformation of engineering in general. What's real, what's important, and where is the most promising activity?

A. Many companies have seen the benefits of digitally transforming their business processes. It has been a key focus for IDC customers and for much of the research we continue to do around digital innovation. Automating manual or repetitive tasks to reduce mistakes and freeing up people power for other, more meaningful work is a key aspect of enabling the creative engagement of teams when restructuring the business for digital transformation and optimization. Developers cite lack of automation as one of the greatest barriers to software development and deployment. The ROI benefits to business processes in this context are well-documented and essential to competitive advantage.

Engineering companies are no exception. These organizations must embrace digitally transforming their engineering processes as well to be competitive. As designs become more digital, they can be optimized faster and cheaper. This optimization enables fast learning cycles and faster convergence to more successful solutions because of the fluidity and adaptability.

We see the use of digital threads to help address these needs. There is also a demand for open, versioned digital threads that encompass visibility across multi-tool environments. Management of information is vital across that web of data, including dependencies and relationships to support impact and coverage analysis for different product versions and variants across environments.

Multidisciplinary teams across the business, development, and operations help bring expertise to the development process. Other keys to success are communication and collaboration, along with effective change management and the use of adaptive, continuous quality processes.

The digital transformation of engineering can be enabled by leveraging and embracing engineering solutions that also enable and benefit from digital thread and digital twin functionality.

Successfully developing increasingly complex products that integrate hardware and software is almost impossible to do without the benefits from digital transformation. Digital transformation not only helps streamline the development of hardware and software systems but also provides a foundation for effective automation of selective tasks across the engineering life cycle. The value of the proposed transformation is magnified by adopting an integrated, end-to-end life-cycle model so that valuable engineering resources are not absorbed in building and maintaining internal tools. Your company's engineers should focus on developing the products you sell. The solution for engineering life-cycle management should provide an end-to-end development model that delivers seamless process handoffs and integration as a core capability.

The consequence of not having these capabilities is grossly underestimated. IDC strongly recommends looking for coordinated, integrated end-to-end engineering management. This can help eliminate wasted people power that's typically spent on maintaining custom code, building interfaces, and migrating data when it's not necessary.

Many organizations still use tools that don't allow them to access dynamic tracking, sharing, updating, and reporting of engineering processes. Engineering teams need to raise their product activity, quality, and compliance while improving

time to market. Effective life-cycle management solutions for engineering can enable new levels of automation, transparency, compliance, and auditability.

It's also helpful to look at what these automated tools have done in the past to enable execution. Make certain that the automated solution you're looking for is flexible and adaptable to the needs of the digital environments in which you're working. You must live with the solution once it's integrated into your development process, so select one with adaptability, longevity, and scalability. That way the system will enable your organization to move forward with consistent product capability.

About the Analyst



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Melinda-Carol Ballou serves as research director for IDC's Application Life-Cycle Management (ALM) program. In this role, Ms. Ballou provides thought leadership, expert opinion and analysis through comprehensive research on end-to-end application life-cycle management – from requirements to quality, testing, change, continuous release, process, project and portfolio management (PPM) with a focus on agile DevOps software life-cycle strategies.

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